# ABT RECERTIFICATION EXAMINATION #24

May 4, 2007 *PART A*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1998 and 2003. There are three parts to the exam. An answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading by **July 2, 2007**. If you wish to return them by courier, the physical address is 1700 Chestnut Hill Road; Wake Forest, NC 27587. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

## EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

## INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

***Good Luck!***

*Key for the following questions*
*Choose the single most appropriate answer*

1. Which of the following is TRUE regarding metal fume fever (MFF)?
   A) MFF resembles polymer fume fever both in its clinical manifestations and causative agents
   B) the causative agents of MFF are dust and fumes of copper and zinc oxides but not of other metals
   C) MFF is characterized by a delay of several hours in the appearance of symptoms
   D) chest X-rays are diagnostic
   E) the mechanism of MFF is a direct toxic effect to the lungs

2. Toxic effects caused by metals include all of the following EXCEPT
   A) blind staggers associated with excessive magnesium intake
   B) degenerative changes in the central nervous system by manganese
   C) respiratory tract carcinogenesis by nickel
   D) peripheral neuropathy and nephropathy by lead
   E) proximal renal tubular effects by cadmium

3. Which of the following is a nonspecific binding protein for metals?
   A) metallothioneins
   B) transferrins
   C) albumin
   D) $B_2$-microglobulin
   E) ferritin

4. *Itai itai* disease is most closely associated with which of the following?
   A) organomercury contamination of fish
   B) glycyrrhizic acid in licorice
   C) cadmium exposure in Japanese women
   D) copper fume exposure in metal workers
   E) lead exposure in children

5. Which of the following organs is/are associated with chronic accumulation and subsequent toxicity of iron?
   A) brain and spinal cord
   B) liver and heart
   C) bone marrow
   D) kidneys
   E) lymph nodes and lymphoid organs

6. Fomepizol may be administered in the treatment of ethylene glycol intoxication to
   A) prevent severe alkalosis
   B) alleviate CNS depression
   C) increase renal excretion of formic acid
   D) increase intracellular NADPH concentrations
   E) inhibit alcohol dehydrogenase

7. Which of the following is ultimately metabolized to oxalic acid?
   A) benzene
   B) propylene glycol
   C) naphthalene
   D) ethylene glycol
   E) sodium hypochlorite

8. Which of the following metals frequently causes a dermatotoxic reaction to jewelry?
   A) aluminum
   B) cobalt
   C) lead
   D) nickel
   E) silver

9. Which of the following is NOT known to be associated with carcinogenic activity?
   A) ethanol
   B) methanol
   C) chloroform
   D) benzene
   E) unleaded gasoline

10. Exposure to which of the following solvents can produce carboxyhemoglobin?
   A) benzene
   B) carbon tetrachloride
   C) ethylene glycol
   D) methyl ethyl ketone
   E) methylene chloride

11. With acute alcohol poisoning in children, it is important to assess
   A) hyperglycemia
   B) poor nutrition status
   C) most recent meal
   D) hypoglycemia
   E) fasting status

12. Which of the following is CORRECT?
A) poisoning from DDT-like compounds and chlorinated cyclodienes involves both sensory and motor components of the central nervous system
B) the extensive skin absorption of DDT-like compounds is responsible for the majority of toxic symptoms seen with this class of insecticides
C) chlorinated cyclodienes are less skin permeable and therefore less acutely toxic than DDT-like compounds
D) carbamate esters are more persistent inhibitors of acetylcholinesterase compared to organophosphate esters
E) the beta and delta isomers of hexachlorocyclohexane are convulsants whereas the alpha and gamma isomers are central nervous system depressants

13. Which of the following is TRUE regarding exposure to acrylonitrile?
A) urinary acrylonitrile and thiocyanate are proposed parameters for biological monitoring of exposed workers
B) brain tumors are a suspected risk in exposed workers
C) bioactivation occurs primarily by CYP2A3
D) parent compound, rather than a metabolite, is likely responsible for the carcinogenicity
E) carcinogenic potency, as determined by $TD_{50}$, is equivalent to that of aflatoxin $B_1$

14. Which of the following is NOT CORRECT about organophosphate-induced delayed neurotoxicity (OPIDN)?
A) OPIDN is caused by some phosphate, phosphonate, and phosphoramidate esters of which only a few have been used as insecticides
B) chickens are immune to organophosphate poisoning and do not display OPIDN
C) the first major epidemic resulted from an alcohol extract of Jamaican ginger contaminated with mixed tolyl phosphate esters
D) inhibition of neurotoxic esterase by >70% generally results in OPIDN
E) signs of poisoning include muscle weakness in the arms and legs giving rise to a clumsy shuffling gait, later replaced by spasticity, hypertonicity, hyper-reflexia, clonus, and abnormal reflexes

15. Which of the following is NOT caused by toxic nerve gases?
A) bronchoconstriction
B) mental confusion
C) increased salivation
D) chloracne
E) bradycardia

16. Which of the following is CORRECT regarding the U.S. Food Quality Protection Act of 1996?
A) identifies children as a subpopulation sensitive to pesticides
B) deals with microbial contamination of the food supply
C) prohibits carcinogenic pesticides in the diet
D) affects only the U.S. Environmental Protection Agency's registration and reregistration processes of pesticides
E) requires organic methods of food production

17. Which of the following is NOT TRUE about parathion?
A) is an organophosphorus insecticide
B) biotransformation of parathion to paraoxon is through oxidative desulfuration catalyzed by the cytochrome P450 whose catalytic activity is increased with ethanol consumption
C) is more toxic than its metabolite paraoxon
D) is one of the pesticides most frequently involved in fatal poisonings
E) may undergo dearylation (detoxification) to yield 4-nitrophenol

18. Which of the following is TRUE of paraquat?
A) it is a highly lipophilic compound that readily crosses cell membranes
B) its toxicity is reduced by decreasing the partial pressure of oxygen
C) its toxicity is limited to the lung
D) it is excreted primarily through the bile
E) its bioactivation by cytochrome P450 enzymes is required for toxicity

19. For a rat nose-only inhalation exposure to vapors, the dose of the test compound can be calculated from
A) the concentration of test compound in the air and body weight of rats
B) the concentration of test compound in the air, body weight of rats and exposure time
C) the concentration of test compound in the air, body weight of rats, exposure time and minute volume
D) the concentration of test compound in the air, body weight of rats, exposure time, and plasma levels of the test compound
E) the volume of the exposure chamber, exposure time and minute volume

20. The biochemical mechanism responsible for death from cyanide poisoning involves
A) formation of methemoglobin
B) formation of carboxyhemoglobin
C) inhibition of cytochrome c
D) inhibition of cytochrome oxidase
E) inhibition of rhodanese

21. Which of the following regarding methylene chloride is TRUE?
- A) large numbers of people are exposed to methylene chloride both occupationally and in the home but systemic toxicity is rare
- B) it produces widespread toxic effects in the liver, kidneys, CNS and heart with high, repeated inhalation exposures
- C) is not carcinogenic in long-term rodent bioassays
- D) does not produce carboxyhemoglobin when inhaled
- E) individuals exposed occupationally commonly experience residual neurologic dysfunction

22. The signs and symptoms included a slight stinging sensation, gradually increasing itching, burning, and redness, followed by a purplish-red area with a white necrotic center and a bluish border, low grade fever, nausea, and vomiting. The size of the necrotic area continued to increase over the next several days, with healing of the injury finally occurring over several weeks. What is the most likely cause of the injury?
- A) allergic contact dermatitis to catechols from the plant *Toxicodendron radicans* (poison ivy)
- B) primary irritant dermatitis caused by contact with an over the counter cosmetic product containing capsaicin
- C) acute dermatitis caused by exposure to histamine, acetylcholine, and serotonin from the hairs of *Urtica urens* (nettles)
- D) *Loxosceles* sp. (brown recluse) spider bite
- E) *Latrodectus* sp. (black widow) spider bite

23. Exposure to excessive quantities of crystalline silicon dioxide dust
- A) causes a toxic response regardless of the crystalline form
- B) causes fibrotic nodules in the lung due to macrophage ingestion of particles
- C) has been definitively shown to cause cancer as a result of the free crystalline silica deposited throughout the interstitium
- D) produces a chronic progressive diffuse interstitial fibrosis with a short latency period
- E) causes a form of pulmonary fibrosis where the lesion is primarily found in the lower zones of the lung

24. Which of the following best describes the respiratory tract deposition of inhaled particles?
- A) deposition by sedimentation is important in the upper respiratory tract
- B) deposition by diffusion occurs as a result of particle inertia
- C) deposition is unaffected by breathing pattern
- D) deposition by interception is important for fibers
- E) ultrafine particles have low deposition in the upper respiratory tract

25. Pulmonary epithelial type I cells
- A) are as metabolically competent as type II cells
- B) constitute a small percentage of the alveolar surface
- C) are precursors to macrophages
- D) are inert to inhaled toxins
- E) produce surfactant

26. Effective, physical and biological half-lives of internally-deposited radionuclides are interrelated such that
- A) physical half-life is always shorter than the effective half-life
- B) the biological half-life is always longer than the physical half life
- C) physical half-life is always longer than the biological half-life
- D) the effective half-life is always longer than the biological half-life
- E) the physical half-life is always longer than the effective half-life

27. Which of the following regarding radon is TRUE?
- A) epidemiology studies show radon and radon daughters cause lung cancer after chronic but not short-term exposure
- B) levels of dust in mines do not affect radon radiation dose delivered to the lung
- C) urban areas tend to have relatively low radon
- D) for the general population, radon exposure is a minor portion of overall radiation dose
- E) the use of open flames in mines can reduce radon dose to lung per unit of exposure

28. Osteogenic sarcoma results from exposure to which of the following radioactive materials?
- A) uranium ($^{235}U$)
- B) radon ($^{222}Rn$)
- C) plutonium ($^{239}Pu$)
- D) radium ($^{224,226,228}Ra$)
- E) iodine ($^{131}I$)

29. All of the following are signs of intoxication in humans associated with alkaloids in blighted or sprouted potatoes EXCEPT
- A) vomiting
- B) diarrhea
- C) debilitation
- D) cardiac stimulation
- E) headache

30. Which of the following is NOT TRUE about *Bacillus anthracis* (anthrax)?
A) can cause cutaneous, inhalation, and intestinal anthrax, with the inhalation form accounting for the most infection cases in many countries
B) anthrax is not known to spread from one person to another person
C) an anthrax vaccine has been licensed for use in humans
D) one of the virulence factors is an antiphagocytic capsular antigen
E) for humans, the source of naturally-acquired anthrax includes contact with infected livestock or contaminated animal products

31. Food hypersensitivity (allergy) is
A) related to a physiologic effect of the food
B) mediated by the immune system
C) never outgrown
D) the same as a metabolic food reaction
E) seen more commonly in adults than in children

32. Which of the following plant chemicals found in belladonna alkaloids acts to block muscarinic receptors and is used for treating patients with significant cholinesterase inhibition in red blood cells?
A) 2,4-diaminobutyric acid
B) neroside
C) atropine
D) ibotenic acid
E) N-oxyalylamino-L-alanine

33. Which of the following is MOST CORRECT concerning food allergies?
A) food allergies are limited to just a few ethnic groups
B) they are believed to be caused primarily by bioengineered food products
C) are only seen in adults
D) are primarily caused by exposure to industrial chemicals in the food supply
E) cutaneous reactions and anaphylaxis are the most common symptoms

34. The pathologic effects of inhaling asbestos fibers
A) progress only when there is long-term continuous exposure to the fibers
B) are independent on the length of fibers
C) do not involve reactive oxygen species
D) may progress to pulmonary fibrosis and malignant mesothelioma
E) are independent of the surface properties of the fibers

35. Which of the following is known to cause a non-immune, food idiosyncratic response?
A) egg whites
B) peanuts
C) green peas
D) fava beans
E) soybeans

36. Aflatoxins are
A) potent neurotoxins
B) relatively nontoxic on an acute basis
C) relatively resistant to biotransformation processes
D) liver carcinogens in certain species only
E) renal carcinogens in humans

37. Aminoglycoside antibiotics have the potential to cause significant toxicity to which of the following organs in both laboratory animals and humans?
A) gastrointestinal tract and liver
B) ovary and uterus
C) bone marrow and spleen
D) inner ear and kidney
E) central nervous system and muscle

38. Which of the following agents inhibits lipolysis of triglycerides, reducing transport of free fatty acids to the liver and therefore decreasing hepatic triglyceride synthesis?
A) cholestyramine
B) niacin
C) clofibrate
D) avasimibe
E) lovastatin

39. Cerivastatin (Baycol), a widely prescribed statin, was removed from the market due to
A) hypercholesterolemia
B) rhabdomyolysis
C) central nervous system stimulation
D) parathesia
E) Torsade de Pointes

40. ICH Guidance "S7A Safety Pharmacology Studies for Human Pharmaceuticals" recommends evaluation of the potential adverse effects of the test substance on the following physiological systems EXCEPT
A) respiratory system
B) autonomic nervous system
C) cardiovascular system
D) gastrointestinal system
E) immune system

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1. Which of the following is TRUE about foreign body tumorigenesis produced by subcutaneously implanted inert plastic or metal films?
   A) multiple perforations in these films increase tumor formation
   B) this process is not considered species specific
   C) tumor initiation is a result of alkylation of DNA
   D) the response is dependent on size and surface area
   E) the chemical nature of the implant is a critical factor

2. Which of the following is a common mechanism by which a compound can induce Leydig cell hyperplasia/adenomas in rodents?
   A) antagonism of dopamine receptors
   B) stimulation of aromatase activity
   C) cholinesterase inhibition
   D) antagonism of gonadotropin releasing hormone receptors
   E) inhibition of 5-alpha-reductase

3. What is the MOST critical factor in determining the region of particle deposition?
   A) lung anatomy and patterns of airflow
   B) particle size
   C) particle shape
   D) impaction and sedimentation
   E) particle nasal clearance

4. What feature does NOT characterize apoptosis?
   A) ordered activation of catabolic processes
   B) chromatin condensation
   C) internucleosomal DNA fragmentation
   D) the formation of apoptotic bodies
   E) increased cell volume

5. An agent is best classified as an animal teratogen when it produces
   A) delayed skeletal development in the fetuses at a dose level that also causes maternal weight loss
   B) visceral malformations in the fetuses at a dose level that also causes maternal weight loss
   C) embryolethality at a dose level that which is lethal to 40% of the maternal animals
   D) malformations at a dose level that has no adverse effect in the maternal animal
   E) abortion and/or early delivery

6. Which of the following is NOT a direct acting genotoxic carcinogen?
   A) nitrogen mustard
   B) bis(chloromethyl)ether
   C) beta-propiolactone
   D) N-methylnitrosourea
   E) benzo[*a*]pyrene

7. In the medium-term bioassay for carcinogenic potential of chemicals, the rodent liver is initially 'programmed' by administration of a necrogenic dose of diethylnitrosamine (DEN), followed by treatment with the test compound and partial hepatectomy. How is the fact that this dose of DEN by itself is carcinogenic overcome?
   A) only rodent strains resistant to DEN-induced carcinogenicity are used
   B) the rodents are first treated with a cytochrome P450 inducer to enhance metabolism of DEN to non-carcinogenic metabolites
   C) DEN-induced tumors are very morphologically distinct and can easily be 'separated' from tumors induced by other chemicals
   D) DEN-induced tumors can easily be differentiated for other chemically-induced tumors by special immunohistochemical staining based on an antibody for DEN-DNA adducts
   E) DEN-induced tumors do not develop in this 8 week bioassay

8. There is evidence of potential adverse developmental effects in both humans and animals from the following chemicals EXCEPT
   A) diphenylhydantoin
   B) diethylstilbestrol
   C) polychlorinated biphenyls
   D) arsenic
   E) organic mercury

9. Which of the following statements about the multi-gene family of glutathione S-transferases (GSTs) is NOT TRUE?
   A) two GSTs, GSTM1 and GSTT1, exhibit common deletion polymorphisms in the human population
   B) GSTs require the co-factor, N-acetylcysteine, for catalytic activity
   C) GSTs play an important role in detoxifying many electrophilic intermediates
   D) most products of GST-mediated conjugation are less toxic than the parent
   E) GSTs are found principally as soluble enzymes in the cell cytosol

10. Which of the following chronic responses to lung injury best matches the mechanism?
A) fibrosis - inhibition of $alpha_1$-antiprotease
B) asthma - hyporeactivity of bronchiolar smooth muscle
C) emphysema - superoxides and free radicals
D) bronchitis - mucous layer thickening
E) lung cancer - changes in collagen cross-linking

11. Which of the following is the most important criteria for a teratogenic response?
A) the ability of a teratogen to bind with low density lipoproteins
B) the pKa of the teratogen at pH 7.4
C) failure of a teratogen to cause maternal toxicity
D) the stage of development at which the embryo is exposed
E) the necessity of the teratogen to cross the placenta

12. A comparison of newborn rats and human infants shows that
A) the blood brain barrier is more developed in the newborn rat than the infant
B) renal function, for example glomerular filtration, continues to develop in both the neonatal rat and infant after birth
C) drug-metabolizing enzymes, especially those involved in conjugation, do not increase until puberty in both the newborn infant and rat
D) neuronal synapses in the central nervous system are complete by birth in the neonatal infant and rat
E) newborn rats and human infants are developmentally similar at birth

13. Which of the following statements is TRUE about glutamate?
A) it is the main excitatory neurotransmitter of the brain
B) it affects axons while having little effect on dendrites and neuronal cell bodies
C) freely diffuses across the blood-brain barrier
D) its effects are mediated by a single, voltage-gated receptor type
E) results in increased resting potential of postsynaptic membranes

14. Which of the following agent-effect pairs is INCORRECT?
A) diethylstilbestrol - genital tract tumors
B) ethanol - low birth weight
C) cocaine - central nervous system effects (including abnormal sleep patterns, tremor, and irritability)
D) retinoids - malformation of face, limbs, and heart
E) valproic acid - phocomelia

15. Silica particles initiate their fibrogenic action on lung tissue because of their recruitment of and cytotoxicity to which of the following?
A) lymphocytes
B) neutrophils
C) macrophages
D) Clara cells
E) fibroblasts

16. Inhalation of beryllium causes chronic lung disease principally through the action of which type of cell?
A) plasma cell
B) monocyte
C) neutrophil
D) T-lymphocyte
E) mast cell

17. Which of the following is NOT associated with human developmental toxicity?
A) polychlorinated biphenyls
B) lead
C) valproic acid
D) chromium
E) methyl mercury

18. Which of the following is NOT reported to produce myelinopathy?
A) triethyltin
B) dinitrobenzene
C) hexachlorophene
D) perhexiline
E) amiodarone

19. The blood-brain barrier
A) is completely developed at birth
B) allows entry of toxic compounds that mimic substrates of active transport systems
C) allows active diffusion of amino acids and neurotoxicants such as MPP+
D) limits the passage of hydrophobic compounds
E) has a remarkable capacity for biotransformation

20. What is the rationale for suggesting the use of mice exhibiting one defective allele of the p53 gene as test subjects for identification of carcinogens?
A) they are resistant to carcinogenesis so false negative responses will be minimized
B) they have a longer life-span than wild-type mice and this maximizes the ability to observe a positive response
C) they have an increased capacity to metabolize genotoxic compounds to inactive metabolites and, therefore, this model is preferred for identifying non-genotoxic carcinogens
D) they have an increased capacity for DNA repair and, therefore, this model is preferred for identifying non-genotoxic carcinogens
E) the p53 gene is a tumor suppressor gene and using mice with a defective allele will make them more susceptible to carcinogenesis

21. The Local Lymph Node Assay
A) is designed to identify agents that are likely to cause systemic oncogenic effects such as lymphoma
B) typically uses guinea pigs as the test system
C) is designed to identify skin-sensitizing chemicals
D) involves oral gavage administration of test materials
E) involves flow cytometry to identify specific T-lymphocyte classes

22. Which of the following is TRUE of functional observational batteries (FOBs) for screening potential neurotoxic agents?
A) include electrophysiological evaluation of nerve function
B) permit the evaluation of a single animal over time to determine the onset, progression, duration and reversibility of a neurotoxic injury
C) exist as two distinct tiers of functional testing, either of which may be selected for testing
D) are useful for evaluating learning and memory
E) comparisons of different protocols of FOBs and motor activity tests suggest that different methods do not identify neurotoxic compounds reliably

23. Which of the following is NOT applicable to the proximal tubules of the kidney?
A) the site of tubular transport of organic anions and cations
B) the most common site of toxicant induced renal injury
C) is susceptible to bioactivated nephrotoxicants
D) has a relatively tight epithelium which impedes the flow of compounds into the cells
E) toxicity is not solely related to accumulation of a nephrotoxicant

24. Which of the following methodologies is probably the best stand-alone assay in rodents for the detection of functional impairment of the immune system?
A) white blood cell count and differential clinical chemistry
B) natural killer cell assay
C) surface marker determination
D) mouse popliteal lymph node assay
E) assessment of the antibody response to a T cell-dependent antigen

25. Which of the following agents is reported to be a photoallergen?
A) penicillin
B) tetrachlorosalicylanilide
C) 8-methoxypsoralen
D) polychlorodibenzofuran
E) sulfonamide

26. Which of the following is NOT a characteristic of the contact urticaria syndrome?
A) may be elicited by non-immunological reactions
B) may be associated with anaphylactic reactions
C) may be IgE mediated reactions
D) is characterized by vasoactive substance release
E) is the same as primary skin irritation

27. The time to final equilibration of the exposure chamber concentration of a gas or vapor is most dependent upon?
A) the specific gas or vapor in use
B) internal surface area of the chamber
C) relative humidity of air in the chamber
D) flow rate and chamber volume
E) none of the above

28. The Draize Test for ocular toxicity involves the evaluation of all of the following EXCEPT
A) corneal opacity
B) depigmentation
C) swelling of conjunctiva
D) reversibility and severity of effects
E) reaction of iris to light

29. The use of inhaled and topical corticosteroids has been associated with which of the following ocular effects?
A) conjunctival hyperemia and decreased ocular pressure
B) myelo-optic neuropathy
C) ocular hypertension and open angle glaucoma
D) retinal swelling and photoreceptor damage
E) corneal band keratopathy

30. Aminoglycoside nephrotoxicity is characterized by all of the following EXCEPT
A) tubular proteinuria
B) enzymuria
C) transport defects resulting in glucosuria and aminoaciduria
D) increased urinary concentrating ability in response to dehydration
E) diminished glomerular filtration rate

31. Which of the following is the MOST useful biochemical index of hepatocellular damage?
A) increased serum alanine aminotransferase
B) decreased serum levels of bilirubin
C) increased inulin clearance
D) decreased hepatic lipid concentration
E) increased serum alkaline phosphatase

32. Which of the following best describes a likely mechanism by which long-term exposure to peroxynitrites in ambient air could contribute to cardiovascular disease?
A) constriction of the coronary vasculature leading to potential myocardial infarction
B) alteration of heart rate or rhythm leading to arrhythmias
C) blockage of K+ channels leading to prolonged action potential and ventricular tachycardia
D) oxidative alteration of low-density lipoproteins leading to formation of atherosclerotic plaques
E) elevation of intracellular Ca+ resulting in cardiac sensitization

33. Hepatic stellate (Ito) cells contribute to hepatotoxic injury because they
A) synthesize collagen during hepatic fibrosis
B) are the major site for vitamin C storage in the body
C) are the major fat-containing cells in fatty livers following hepatic injury
D) are the resident macrophages of the liver
E) are the major source of hepatic lipid used in bile formation

34. In a 3-month study in rats, a chemical caused both centrilobular hepatocellular hypertrophy and thyroid follicular cell hyperplasia. Which of the following is the MOST likely explanation for these observations?
A) induction of hepatic microsomal enzymes with resultant increased biliary excretion of thyroxin (T4) with a subsequent compensatory decreased output of thyroid stimulating hormone (TSH) by the pituitary
B) induction of hepatic microsomal enzymes with resultant increased biliary excretion of T4 and a subsequent compensatory increased output of TSH
C) chronic inhibition of hepatic microsomal degradation of triiothyronine (T3) with a subsequent compensatory decreased output of TSH by the pituitary
D) stimulation of hepatic microsomal degradation of TSH resulting in lowered plasma concentrations of TSH
E) direct mitogenic activity of the test compound on the thyroid follicular cells

35. Recent studies have shown that 1, 3-butadiene increases the incidence of cardiac hemangiosarcomas, tumors originating in which of the following cell types?
A) canalicular cells
B) endothelial cells
C) smooth muscle cells
D) foam cells
E) myocytes

36. Renal papillary necrosis with chronic nephritis is generally associated with
A) chronic alcohol abuse and aspirin use
B) chronic excessive consumption of nonsteroidal anti-inflammatory agents (NSAIDs)
C) chronic excessive consumption of fish containing mercury and chronic excessive abuse of opioid analgesics
D) chronic caffeine use, alcohol abuse and cigarette smoking
E) chronic diethylene glycol exposure and chronic NSAIDs abuse

37. The kidneys play a principal role in all of the following EXCEPT
A) acid-base balance
B) electrolyte composition
C) regulation of extracellular fluid balance
D) excretion of protein bound materials
E) excretion of urea

38. Proliferation of peroxisomes in mammalian hepatocytes
   A) is a nonspecific effect that is not receptor mediated
   B) can be detected by histochemical staining for catalase
   C) generally results in a preferential increase of fatty acid beta oxidation activity over catalase activity
   D) is associated with DNA damage
   E) is of mechanistic importance in human liver carcinogenesis

39. An increase in which of the following serum enzymes best reflects impaired biliary excretion (cholestasis) as opposed to parenchymal necrosis?
   A) alanine transaminase
   B) sorbitol dehydrogenase
   C) amylase
   D) alkaline phosphatase
   E) lactate dehydrogenase

*Key for the following questions*
*Choose A) if 1,2 and 3 are correct*
*Choose B) if 1 and 3 are correct*
*Choose C) if 2 and 4 are correct*
*Choose D) if only 4 is correct*
*Choose E) if all four answers are correct*

40. Potential adverse effects of bone marrow transplant in combination with chronic immunosuppressive treatment with cyclophosphamide in cancer patients include
   1) graft versus host disease
   2) cytomegalovirus infection
   3) *Candida albicans* infection
   4) leukemia

# ABT RECERTIFICATION EXAMINATION #24

May 4, 2007 *PART C*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1998 and 2003. There are three parts to the exam. An answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading by **July 2, 2007**. If you wish to return them by courier, the physical address is 1700 Chestnut Hill Road; Wake Forest, NC 27587. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

## EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

## INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

***Good Luck!***

*Key for the following questions*
*Choose the single most appropriate answer*

1. Which of the following group of organs has a major role in the biotransformation of xenobiotics?
   A) liver, kidneys, skin, spleen and lungs
   B) liver, brain, gastrointestinal tract and skin
   C) liver, adrenal glands, brain, skin and lungs
   D) liver, kidneys, gastrointestinal tract, skin and lungs
   E) liver, heart, gastrointestinal tract, skin and lungs

2. Xenobiotics that cause the uncoupling of the mitochondrial respiratory electron transport chain are particularly toxic because
   A) electron flow is maintained but no ATP is produced
   B) heat is generated via thermogenesis
   C) $Ca^{+2}$ flux out of the mitochondrial matrix occurs
   D) the mitochondrial membrane potential is increased
   E) the electron transport flow is reversed

3. Alteration of gastrointestinal motility may be manifested in one of two ways, the MOST important of which is
   A) decrease in intestinal motility leading to constipation and irritated bowel syndrome
   B) decrease in intestinal motility leading to a decrease in absorption of nutrients
   C) decrease in intestinal motility leading to frequent defection
   D) increase in intestinal motility leading to a decrease in absorption of nutrients
   E) increase in intestinal motility leading to frequent defecation

4. Binding of xenobiotics to plasma proteins by reversible hydrogen bonding is of special interest to toxicologists because
   A) agents bound to plasma proteins may significantly alter the buffering capacity of plasma
   B) this frequently results in immunotoxicity by antigen-antibody reactions
   C) only certain chemical classes are bound to plasma proteins to any significant degree
   D) displacement of the compound from protein binding sites by another agent can alter the systemic exposure
   E) binding tends to occur mainly to plasma globulins and may thereby alter functional antibody titers

5. Half-life of elimination is
   A) the time required for the concentration of the compound in blood or plasma to decrease by one-half
   B) independent of the volume of distribution
   C) independent of clearance (C1)
   D) not constant for compounds eliminated by first-order kinetics
   E) calculated by the following formula: (0.693 x C1) $V_d$

6. Polychlorinated biphenyls (PCBs) are of potential concern to human health because of all of the following EXCEPT
   A) they have been shown to have neurological and reproductive effects in animal models
   B) many people are exposed to PCBs because the main route of exposure is dietary intake
   C) certain PCBs bind to the aryl hydrocarbon receptor and induce toxicological effects similar to those of dioxins and dibenzofurans
   D) they are widespread in the environment and can undergo global transport and dispersion
   E) they are not metabolized and not cleared from the body in humans

7. Which of the following is NOT TRUE of occupational asthma?
   A) occupational asthma is the most common occupational pulmonary disease in developed countries
   B) a decline in $FEV_1$ of 20% over a workday is highly suggestive of occupational asthma
   C) a progressive decline in $FEV_1$ over the work week is highly suggestive of occupational asthma
   D) a characteristic feature is airway hyporesponsiveness to stimuli in the occupational setting
   E) highest prevalence is seen in current or former smokers

8. Which of the following BEST describes a suicide inhibitor of the microsomal cytochrome P450 metabolizing enzyme system?
   A) causes the reversible, competitive inhibition of a specific cytochrome P450
   B) only affects rodent liver due to the abundance of peroxisome proliferator-activated receptor alpha (PPARa)
   C) forms a reactive metabolite that subsequently binds to the active site with higher affinity than the parent xenobiotic
   D) does not inhibit the same P450 enzyme that transforms it to a reactive metabolite
   E) blocks the binding of a true substrate at the active site

9. Which of the following tissue/organ compartments by itself is NOT likely to be included in a physiologically-based toxicokinetics model for an inhaled volatile organic chemical?
A) liver
B) fat
C) lung
D) intestines
E) richly-perfused tissues

10. A chemical is eliminated by zero-order kinetics, and 20% of an ingested dose is eliminated in the first hour. At what time point after ingestion would 30 ml of the original dose remain in the body of the original dose was 50 ml?
A) 2 hour
B) 2.5 hour
C) 3 hour
D) 4 hour
E) 5 hour

11. The initial step in the conversion of 1,2,3,6-tetrahydro-1-methyl-4-phenylpyridine (MPTP) to 1-methyl-4-phenylpyridine ($MPP^+$) is
A) epoxidation by a cytochrome P450
B) N-acetylation to produce a reactive nitrenium ion
C) metabolism by prostaglandin H synthase to a carbon centered radical
D) glucuronidation to a reactive acyl glucuronide
E) metabolism by monoamine oxide in astrocytes

12. The value of apoptosis as a tissue repair strategy is markedly diminished in which of the following cell types?
A) bone marrow
B) gastrointestinal epithelium
C) cardiac muscle cells
D) skin epidermis
E) hepatic parenchymal cells

13. The genetic condition known as "Fish Odor Syndrome", where failure to fully metabolize certain protein components of the diet (for example, choline, lecithin, carnitine) results in excretion of the highly odorous compound, trimethylamine, is associated with a deficiency in
A) Cytochrome P450 2D6 (CYP2D6)
B) Glutathione S-transferase T1 (GSTT1)
C) N-acetyl-transferase 2 (NAT2)
D) flavin-dependent monooxygenase 3 (FMO3)
E) UDP-glucuronosyl transferase 1A1(UGT1A1)

14. Which of the following is TRUE concerning trichloroethylene (TCE)?
A) induces liver, lung and kidney tumors in rats and mice
B) human kidney tumors in exposed individuals are associated with mutations in the von Hippel-Lindau tumor suppressor gene
C) from biopsy specimens, human liver tumors in exposed individuals display an excess of peroxisomes
D) the low levels of glutathione in the lung limit conjugation and excretion of the TCE metabolite chloral
E) Type 2 cells in the mouse lung are the target of TCE toxicity

15. A general mechanism for detoxification of electrophiles is
A) conversion by super oxide dismutase to HOOH
B) conjugation with glutathione
C) reduction by NADPH-dependent flavoenzymes
D) oxidation by monooxygenases
E) peroxide-catalyzed conversion to free radicals

16. Which of the following BEST describes hormesis for non-nutritional xenobiotics?
A) toxicity remains the same at both low and high doses
B) toxicity has no dose relationship
C) a beneficial or stimulatory effect at low doses and toxicity at high doses
D) no threshold for toxicity
E) the dose response is monotonic

17. Bioavailability is assumed to be 100% for which of the following routes of exposure?
A) inhalation
B) dermal
C) intravenous
D) subcutaneous
E) intranasal

18. The antihistamine terfenadine has been associated with *torsades de pointes* (a cardiac arrhythmia) due to
A) inhibition of the rapidly-activating delayed rectifier current, $I_{Kr}$, with resultant prolongation of the QT interval
B) inhibition of cytochrome P450
C) activation of the rapidly-activating delayed rectifier current, $I_{Kr}$, with resultant prolongation of the QRS interval
D) blockage of $Ca^{2+}$ channels with a resultant slowing of the heart rate and hypotension
E) antagonism of beta-adrenergic receptors, increased PR interval and a slowing of the heart rate

19. Which of the following does NOT undergo biotransformation to a toxic metabolite?
A) TCDD (2,3,7,8-tetrachlorodibenzodioxin)
B) acetaminophen
C) chloroform
D) hexane
E) ethylene glycol

20. Induction of which of the following families of cytochrome enzymes is associated with the mode of action of hepatocellular tumor induction with di-(2-ethylhexyl)phthalate?
A) CYP1A
B) CYP2A
C) CYP2B
D) CYP3A
E) CYP4A

21. Epidemiological cross-sectional studies are also known as
A) prospective studies
B) prevalence studies
C) case-control studies
D) cohort studies
E) transectional studies

22. The U.S. Food and Drug Administration recommends which one of the following sequences in order to market a new pharmaceutical?
A) preclinical testing, followed by submission of a new drug application (NDA), then clinical trials, and finally, submission of an investigational new drug (IND) application for exemption
B) submission of an NDA followed by preclinical testing, then clinical trials, and finally, submission of an IND application
C) preclinical testing, followed by submission of an IND application, then clinical trials, and finally, submission of an NDA
D) submission of an IND application, then clinical trials, and finally, submission of an NDA
E) preclinical testing, then clinical trials, followed by submission of an NDA

23. Which of the following is relevant for a Threshold Limit Value (TLV)?
A) most workers may be exposed safely to this concentration
B) the exposure period is 24 hours per day
C) the exposure frequency is seven days per week
D) the exposure duration is 70 years
E) enforced by the U.S. National Institute for Occupational Safety and Health

24. What would be the MOST critical objective in conducting toxicology studies in support of the commencement of human clinical trials in healthy volunteers?
A) to identify potential toxicities for monitoring in humans
B) to provide a basis to select a safe clinical starting dose
C) to correlate animal toxicities to toxicities that may be found in humans
D) because toxicology studies are required by the Food and Drug Administration before the commencement of human clinical trials
E) to establish an NOAEL

25. The U.S. Environmental Protection Agency establishes an inhalation reference concentration (RfC) for non-cancer hazards posed by air pollutants. The RfC is defined as
A) the no observed adverse effect level (NOAEL)
B) the concentration known to impose no risk to the maximally exposed individual
C) an estimate of the concentration that is likely to be without appreciable risk to the exposed population after continuous lifetime exposure
D) below detectable limits
E) the concentration estimated to impose a risk of harmful effects of less than $10^{-5}$ to the maximally exposed population

26. Good Laboratory Practices
A) ensure that appropriate study design is used
B) differ depending on the test article and the regulatory agency
C) provide an oversight role for laboratory management
D) are required for range-finding and exploratory studies
E) allow inspection, but not disqualification, of laboratories

27. The Delaney Clause (of the U.S. Food, Drug and Cosmetic Act, as amended in 1958) essentially prohibits food additives that have been shown in laboratory animals or humans to cause which of the following adverse effects?
A) cancer
B) mutations
C) birth defects
D) anti-cholinesterase activity
E) xenoestrogen activity

28. The Tg-AC transgenic mouse carcinogenicity model
A) is approved by ICH to evaluate carcinogenic potential of all pharmaceutical classes
B) is considered appropriate for both genotoxic and non-genotoxic compounds
C) is considered useful for screening the carcinogenic potential of dermally administered pharmaceuticals
D) reacts consistently and completely to known human carcinogens
E) is equally responsive to topically and systemically administered compounds

29. Which of the following is a statistical tool useful for analyzing toxicology data?
A) Fisher's Exact Test for the analysis of three or more sets of data
B) Chi Square Test for the analysis of two sets of discontinuous data
C) F Test, which is typically the initial step in comparing two groups of continuous parametric data
D) ANOVA for the comparison of three or more groups of continuous data where the variances are heterogeneous and the data are not normally distributed
E) nonlinear regression when the change in one variable is directly proportional to a change in another variable

30. Epidemiologic case-control studies
A) are also known as prospective studies
B) can be used to calculate "risk factors"
C) specifically ignore demographic variables such as age and sex
D) often take a decade or longer to complete the data collection
E) are also known as randomized controlled trials

31. A 2 year-old child is rushed to the emergency room, having swallowed a sodium hydroxide-based household cleaning product. Which of the following is indicated as immediate treatment if the child is conscious?
A) have the child drink lemon juice or vinegar to dilute the product and decrease the pH in the gastrointestinal tract
B) induce vomiting
C) rinse any visible material from the child's mouth with water or saline
D) give activated charcoal orally
E) give an enema

32. Which of the following treatments is MOST likely to be beneficial in the case of antifreeze ingestion?
A) ethanol infusion
B) emetic to induce vomiting
C) removal of stomach contents
D) administration of activated charcoal
E) administration of N-acetylcysteine

33. A warning is present on acetaminophen labels stating not to consume excessive amounts of ethanol while using this drug. What is the basis for this warning?
A) both compounds are respiratory depressants
B) both compounds are gastrointestinal irritants
C) both compounds are metabolized by cytochrome P450 2E1; therefore, acetaminophen decreases the metabolism and increases the toxic effects of ethanol
D) ethanol induces cytochrome P450 2E1 which converts acetaminophen to a toxic metabolite
E) more acetaminophen is absorbed after ethanol consumption

34. Which of the following concerning biodegradation of chemicals in the environment is TRUE?
A) biodegradation occurs only under aerobic conditions
B) higher organisms play the most important role in biodegradation because they are able to metabolize compounds more efficiently
C) all chemicals that are biodegradable are usable by the organism immediately as a nutrient
D) microorganisms are the primary converters of complex organic chemicals to inorganic substances
E) biodegradation occurs only in the presence of light

35. Which of the following air pollutants is emitted in the greatest abundance in the U.S.?
A) nitrogen oxides
B) carbon monoxide
C) particulate matter
D) sulfur oxides
E) hydrocarbons

36. The following factors, identified by Bradford-Hill, must be apparent for establishing a causal relationship EXCEPT
A) consistency of the association
B) strength of the association
C) specificity of the association
D) coherence or biological plausibility
E) statistical association between study groups

37. The probability that a chemical may produce cancer in humans can best be estimated by knowing its dose response relationship from an animal bioassay and
- A) whether comparable peak plasma concentrations are achieved in humans and rodents
- B) the magnitude and duration of human exposure
- C) whether the chemical induces oncogenes such as p53
- D) whether the chemical causes cytotoxicity
- E) if the chemical induces oxidative stress

38. Elevation of which of the following serum/plasma biomarkers is MOST indicative of skeletal muscle toxicity?
- A) troponin I
- B) alkaline phosphatase
- C) creatine kinase
- D) aspartate aminotransferase
- E) creatinine

39. Short Term Exposure Limits (STELs) are
- A) the absolute maximum concentrations to which a person should be exposed
- B) higher than TLV-ceiling levels
- C) levels to which most people can be exposed for up to 15 minutes continuously without adverse effects
- D) exposures that are tolerable during an 8-hour workday
- E) intermittent exposure levels set by the US Food and Drug Administration and US Environmental Protection Agency

40. Which of the following is NOT a correct pairing of occupational disease and the agent that causes it?
- A) hepatic steatosis (fatty liver) - carbon tetrachloride
- B) Legionellosis - *Borrelia burgdorferi*
- C) chloracne - polychorinated biphenyls
- D) hepatic hemangiosarcoma - vinyl chloride
- E) chronic bronchitis - cotton dust